Unsupervised discovery via β-VAE · 84k patients · 2003–2020

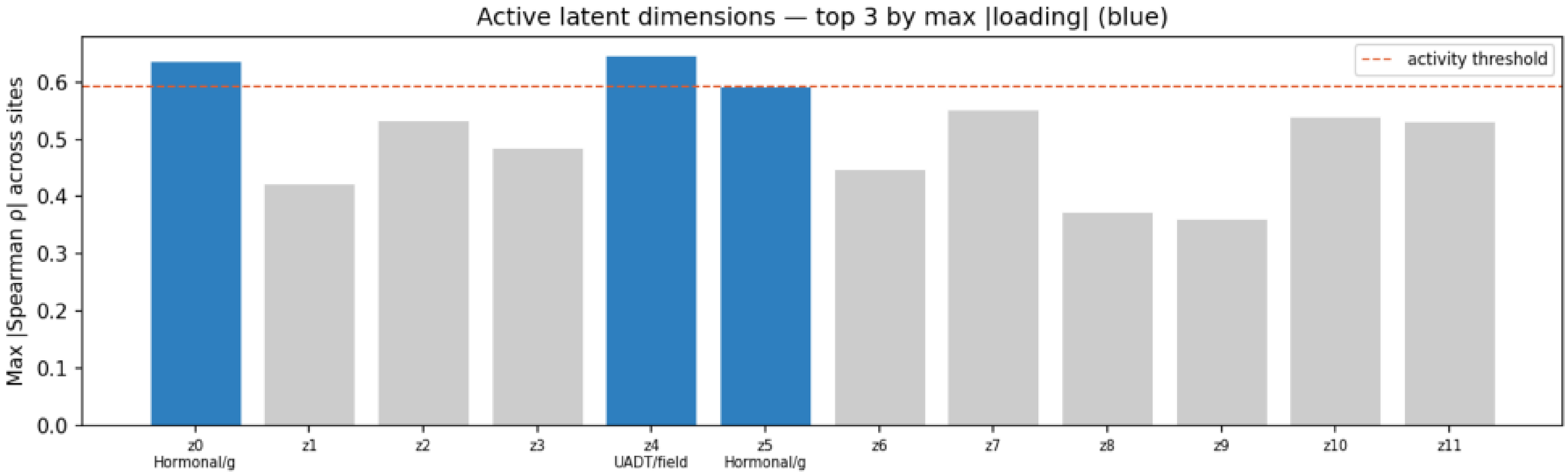
Cohort: 78,442 patients · Multi-cancer: 4,052 (5.2%)

VAE latent space: 12 dimensions → 3 active axes (highest max |loading|)
KMeans clustering: k=5, silhouette=0.529

Key finding: Cancer co-occurrence in Taiwan is explained by three latent axes, corresponding to distinct carcinogenic exposures (UADT/field-cancerization, hormonal/gynecologic ×2). Cluster survival differs significantly (log-rank $p<0.001$).
NOTE: proportional hazards violated for 5/6 model terms; time-split Cox (landmark=2yr) used as primary analysis — full-follow-up HRs are invalid.

Active Latent Axis Identification

Fig 1: Max |Spearman ρ| per latent dimension — top 3 (blue) selected as active axes

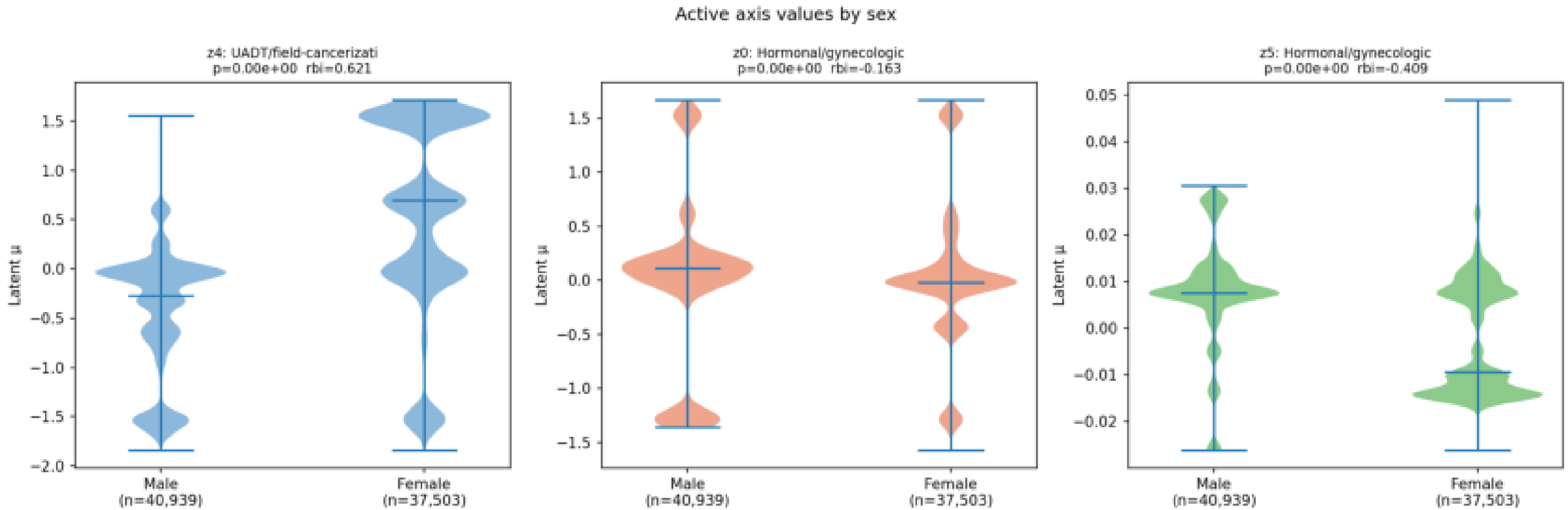


Axis interpretation (all 12 dims; active in bold)

Dim	Axis name	Top sites (+)	Top sites (–)	Max |ρ|
z0	Hormonal/gynecologic	C18, C20, C61	C22, C53, C50	0.637
z1	UADT/field-cancerization	C34, C61, C20	C16, C06, C02	0.422
z2	Hormonal/gynecologic	C22, C67, C02	C50, C53, C34	0.534
z3	Liver-GI	C18, C20, C67	C22, C06, C50	0.485
z4	UADT/field-cancerization	C50, C53, C20	C34, C06, C02	0.647
z5	Hormonal/gynecologic	C61, C18, C20	C50, C06, C53	0.593
z6	Liver-GI	C34, C20, C06	C53, C22, C42	0.448
z7	UADT/field-cancerization	C34, C22, C61	C50, C06, C02	0.551
z8	Hormonal/gynecologic	C53, C20, C50	C18, C22, C15	0.374
z9	UADT/field-cancerization	C34, C06, C02	C18, C61, C67	0.362
z10	Liver-GI	C22, C61, C34	C53, C20, C42	0.54
z11	Hormonal/gynecologic	C18, C20, C53	C22, C06, C54	0.532

Axis Characterisation — Sex

Fig 2: Active axis values by sex (violin plots, Mann–Whitney U)



z4 [UADT/field-cancerization]: sex MWU p=0.00e+00 rbi=0.621 | age ρ=-0.232 p=0.00e+00 | era ρ=0.003 p=3.48e-01
z0 [Hormonal/gynecologic]: sex MWU p=0.00e+00 rbi=-0.163 | age ρ=0.113 p=7.66e-221 | era ρ=0.108 p=4.28e-201
z5 [Hormonal/gynecologic]: sex MWU p=0.00e+00 rbi=-0.409 | age ρ=0.333 p=0.00e+00 | era ρ=0.070 p=1.00e-85

Axis Characterisation — Age and Temporal Trend

Fig 3a: Active axes vs age at first diagnosis

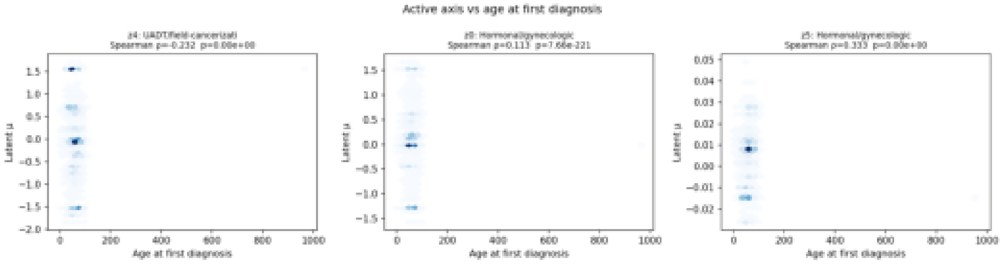
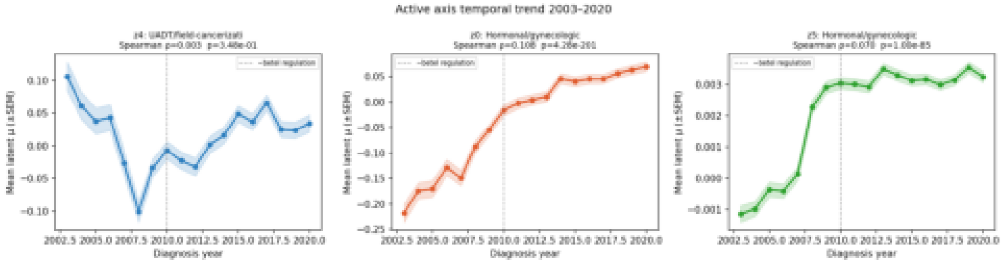


Fig 3b: Active axis temporal trend 2003–2020
(vertical dashed: ~2010 betel nut regulation)



Cluster Structure

Fig 4a: Silhouette analysis k=2–8
(k=8 best; k=5 selected; sil=0.529)

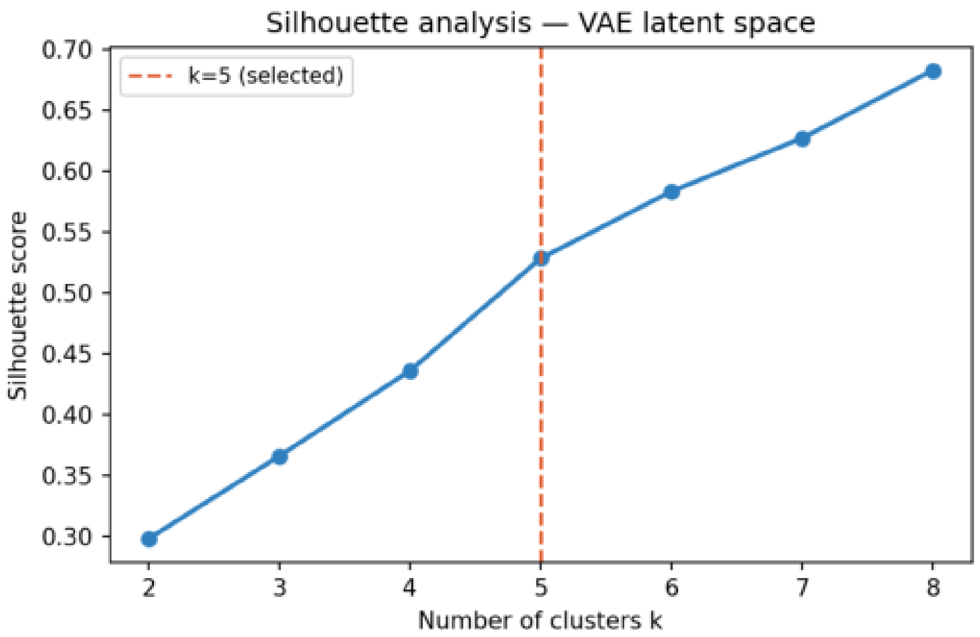
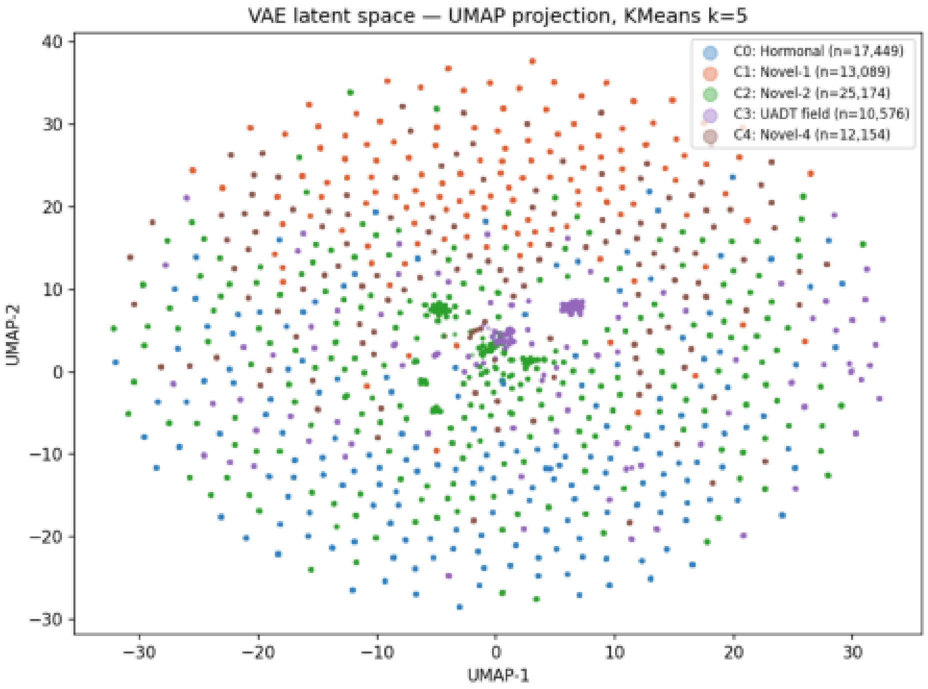


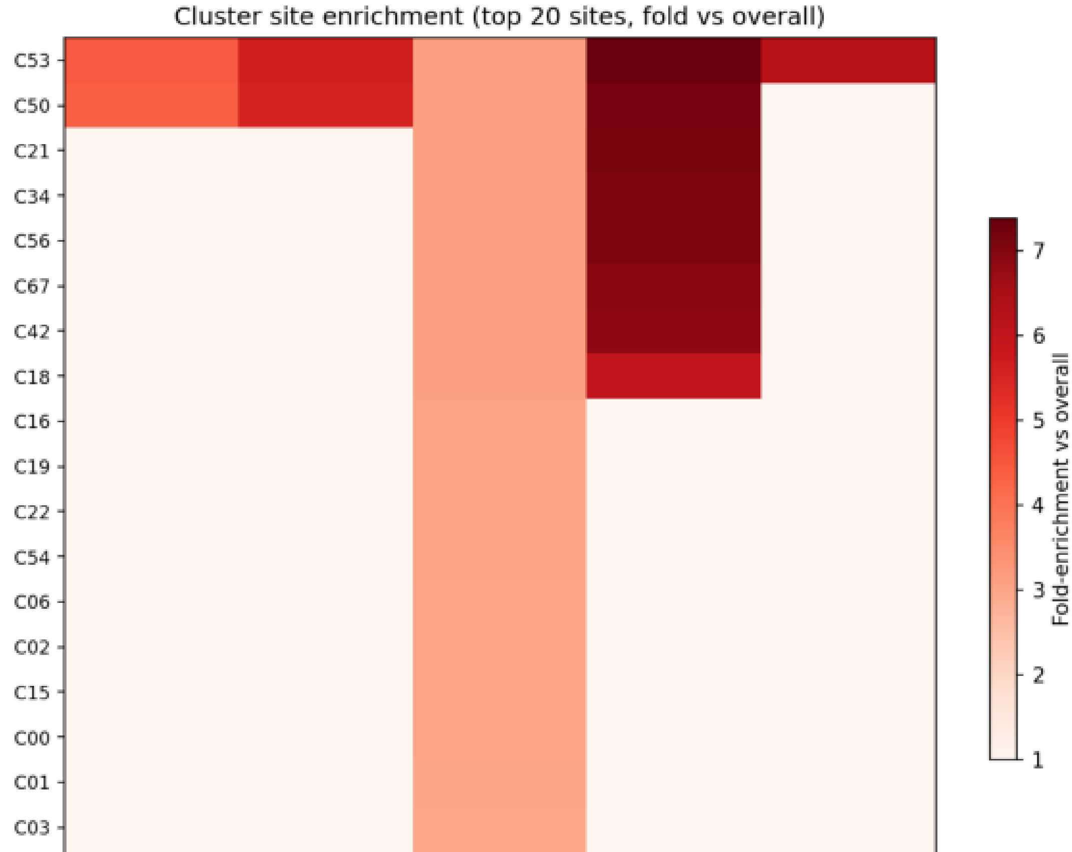
Fig 4b: UMAP of VAE latent μ — KMeans k=5 clusters



C0: Hormonal (n=17,449) C1: Novel-1 (n=13,089) C2: Novel-2 (n=25,174) C3: UADT field (n=10,576) C4: Novel-4 (n=12,154)

Cluster Site Enrichment

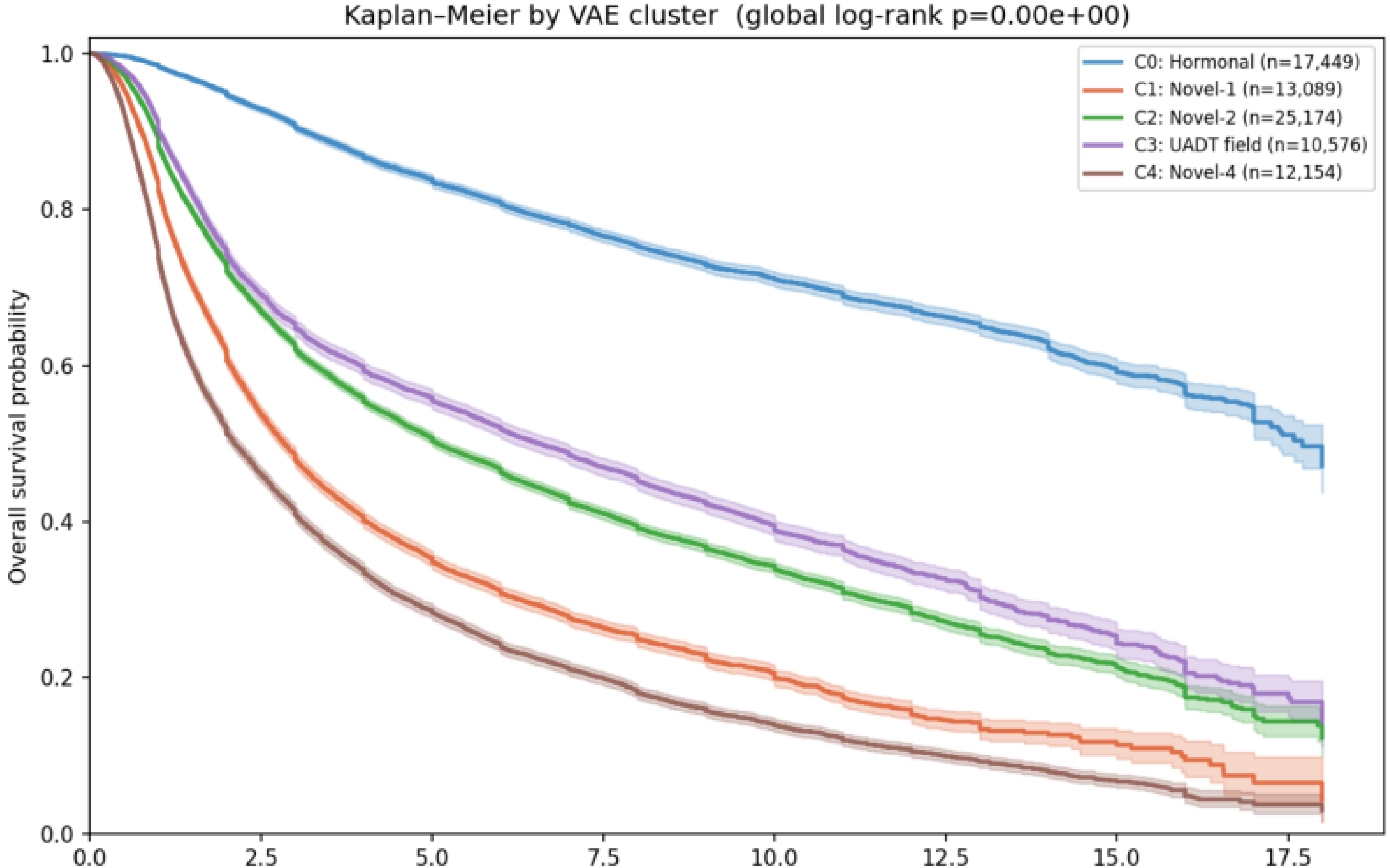
Fig 5: Cluster × site fold-enrichment heatmap (top 20 sites, fold vs overall mean)



Cluster	Name	N	Top-5 enriched sites
0	Hormonal	17449	C53, C50, C21, C34, C56
1	Novel-1	13089	C34, C61, C71, C22, C67
2	Novel-2	25174	C20, C77, C44, C38, C30
3	UADT field	10576	C54, C56, C06, C16, C02
4	Novel-4	12154	C22, C08, C69, C17, C21

Survival Impact of Cluster Membership

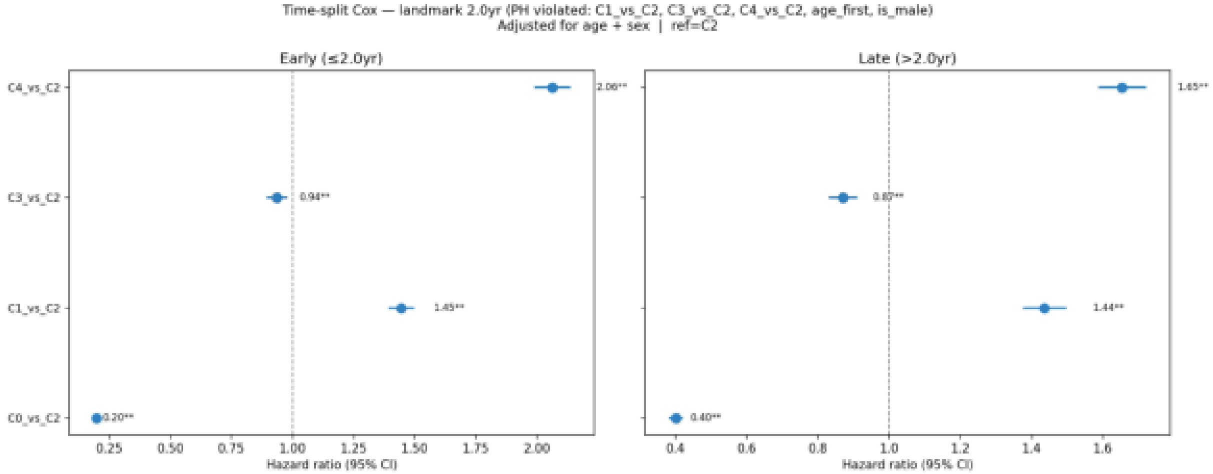
Fig 6: Kaplan–Meier overall survival by VAE cluster
(global log-rank test; time from first diagnosis, approximate)



C0 [Hormonal]: n=17,449 median=5.0yr dead=3706 C1 [Novel-1]: n=13,089 median=2.0yr dead=8184 C2 [Novel-2]: n=25,174 median=2.8yr dead=12891 C3 [UADT field]: n=10,576 median=3.0yr dead=5118 C4 [Novel-4]: n=12,154 median=1.8yr dead=9320

Survival Analysis — Time-Split Cox (Primary) + PH Test

Fig 7 (PRIMARY): Time-split Cox — landmark 2yr
PH assumption violated for C1/C3/C4/age/sex — full-FU model invalid



Time-split Cox (landmark=2yr):
C0: HR=0.20 early / 0.40 late
C1: HR=1.45 early / 1.44 late
C3: HR=0.94 early / 0.87 late
C4: HR=2.06 early / 1.65 late

Interpretation:

- C3 (Mixed-GI/gynecologic) HR=44 in full model is ARTIFACTUAL — PH violation confirmed (χ^2=177.6, p=10^{-40}). Time-split shows C3 is PROTECTIVE vs C2 in both periods.
- C0 (Hormonal) satisfies PH — strongly protective vs reference throughout follow-up.
- C1/C4 show time-varying attenuation (early excess hazard diminishes over time).

Limitations:

① PH violated for 5/6 terms — time-split Cox used as primary; full-FU HRs not reported.
② k=5 pre-specified; silhouette monotone → k=8.
③ Survival time approximated from diag_yr Jan 1.
④ Cluster labels require manual ICD-O annotation.
⑤ No molecular/treatment data in registry.

Schoenfeld PH test | Next steps: ICD-O cluster annotation; β sensitivity; held-out validation

	χ^2	p	PH OK
C0 vs C2	0.3	5.8e-01	✓ Yes
C1 vs C2	317.7	4.5e-71	✗ No
C3 vs C2	177.6	1.6e-40	✗ No
C4 vs C2	1093.7	7.6e-240	✗ No
age_first	273.3	2.1e-61	✗ No
is_male	119.7	7.2e-28	✗ No

Next steps: ① Manual ICD-O annotation of Novel-1/Novel-2. ② HBV era-split (pre/post 1986). ③ β sensitivity (β=2,4). ④ Held-out 20% validation. ⑤ Cluster-conditioned Transformer (Script 07).